

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 2.4 ACIDS, BASES AND SALTS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Look again at the two images from Lesson 1: (1) Lake Magadi with its white trona crusts and pink-red alkaline water, and (2) the aluminium sufuria with white scale inside and pitting outside where lemon-water sat overnight. Use everything you learned in Lessons 1 to 8 to answer each question in the space under it. Write in full sentences. Use the Claim–Evidence–Reasoning (CER) pattern where the question asks for it: state your claim, give evidence from our investigations, and explain your reasoning using ions, particles or reactions. Include equations or sketches where they help your answer. Use correct chemical vocabulary and Kenyan examples from our lessons.

Part 1: Why is Lake Magadi water caustic, and why does it form white crusts?

Lake Magadi water looks dangerously caustic, yet the Maasai have harvested its salts for centuries and the Magadi Soda Company makes soda ash from it. Using the Claim–Evidence–Reasoning pattern, explain (a) why the water is caustic, in terms of pH and ions, and (b) how the white trona crusts form on the shoreline. Name the salt in the crusts and the product made from it.

Claim: Lake Magadi water is caustic because it is a strongly basic solution, and the white crusts form when the water evaporates and the dissolved salt crystallises. Evidence: In our pH investigation (Lesson 2) the simulated Lake Magadi water turned universal indicator to the basic range, pH 9–11, and wood-ash solution also tested basic. In Lesson 3 we saw that the lake's pH is about 10.5, similar to bleach, which is alkaline. Reasoning: Dissolved trona (sodium sesquicarbonate, $\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$) is a sodium carbonate salt that releases excess hydroxide ions (OH^-) in water. A high concentration of OH^- ions means a high pH, and these ions react with the proteins and oils of unprotected skin, which is why the water is caustic. The Rift Valley heat evaporates the water, so the dissolved salt becomes more concentrated and crystallises as white crusts on the shore. The crusts form by crystallisation as the water dries. They are the raw material for soda ash (Na_2CO_3), which is useful as a cleaning and bleaching agent and is the anhydrous form of washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$). So the same alkaline salt is caustic and useful, depending on how it is used and how concentrated it is.

Part 2: Why did lemon juice pit the sufuria?

Lemon juice is safe to drink, yet lemon-water left overnight pitted the outside of the aluminium sufuria. Explain this using the pH of lemon juice, the ions involved, and what happened when we reacted an

Lemon juice is acidic, with a pH of about 2–3, because its citric acid releases hydrogen ions (H^+) in water. In Lesson 4 we reacted dilute hydrochloric acid with magnesium ribbon, our model metal for aluminium. The ribbon disappeared with rapid fizzing, and the glowing splint test gave a squeaky pop, showing hydrogen gas. The word equation is: acid + metal $\rightarrow$ salt + hydrogen. In the sufuria, the H^+ ions in the lemon-water reacted with aluminium metal overnight, forming an aluminium salt and hydrogen gas. This removed metal from the surface and left grey pits. Drinking diluted lemon juice is safe because it is dilute and the stomach lining is protected by mucus, so it is not attacked the way bare metal is. Our body also has no bare metal surface to corrode. Acids are not simply dangerous. How much acid there is, and what it touches, decide whether it is useful or harmful.

acid with a metal in Lesson 4. Write a word equation for the type of reaction. Then explain why the same lemon juice does not damage your body when you drink it.	
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Part 3: Why does salt clump, and where did the white scale come from?

(a) A school cook in Mombasa finds her bulk salt clumps into a block in the rainy season. Using our petri dish results for NaCl, CaCl_2 , $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, explain why this happens. Use the terms hygroscopic, deliquescent and efflorescent correctly. (b) The white scale inside the sufuria is also a salt. Explain how it formed, and say how it differs from the trona at Lake Magadi.	(a) Salts interact differently with water vapour in the air. In our 24-hour petri dish investigation, anhydrous CaCl_2 gained mass and partly dissolved into a liquid, so it is deliquescent: it absorbs so much water vapour that it dissolves. $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ lost mass and turned powdery, so it is efflorescent: it loses water of crystallisation to the air. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ showed slight mass loss. NaCl showed very little change, so it is only weakly hygroscopic, meaning it takes in small amounts of moisture without dissolving. Na^+ and Cl^- ions attract polar water molecules from the air. Over weeks of very high coastal humidity, enough water is absorbed to make the grains stick together and clump. (b) The scale inside the sufuria is left by hard water, which contains dissolved mineral salts. When the water evaporates or boils off, the dissolved salt is left behind as a solid scale. The scale is calcium carbonate, which is a different salt from the sodium carbonate trona at Lake Magadi. Both are salts that crystallise when water evaporates.
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Part 4: Salts in our lives and the environment

Answer both parts. (a) Choose TWO different sectors from Lesson 6 (agriculture, food, medicine, laundry, road use). For each, name a specific salt and explain how its chemistry makes it useful. (b) Using the Lake Victoria case study from Lesson 8, explain how using too much fertiliser salt can harm the lake and its fish. Then answer: how can understanding salt chemistry help us use salts safely and sustainably?	(a) Medicine: antacids contain basic salts such as CaCO_3 and $\text{Mg}(\text{OH})_2$. They neutralise excess stomach hydrochloric acid, and carbonate antacids fizz because acid + carbonate $\rightarrow$ salt + water + carbon dioxide. Laundry: washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) has a high carbonate ion concentration and softens hard water by precipitating Ca^{2+} and Mg^{2+} ions. It is linked to the soda ash from Lake Magadi trona. (Other correct sectors: CAN and DAP supply ammonium and nitrate ions that plants need to make protein; NaCl preserves nyama choma and fish; NaCl gritting lowers the freezing point of water on highland roads.) (b) CAN and DAP are soluble salts. When too much is applied, rain washes the dissolved nitrate and phosphate ions into Lake Victoria. The case study showed higher nitrate values in the rainy season at Kisumu Port, Dunga Beach and Kendu Bay. The extra nutrients cause eutrophication: algae and water hyacinth grow explosively, and when this biomass decomposes it uses up dissolved oxygen, so fish such as Nile tilapia (ngege) suffocate. Safe and sustainable use means knowing the properties of each salt: apply fertiliser in the right amounts, limit dietary NaCl because excess raises blood pressure, and handle acidic and basic substances at the right concentration. The same salt that is essential in the right amount can be harmful in excess.
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Part 5: Your model has changed

Think back to your Lesson 1	(a) Sample answer: In Lesson 1 I thought all salts were the same as table salt and that anything that tastes sour or burns must be
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two-panel sketch and predictions. (a) Describe one idea you held in Lesson 1 that you have now changed or improved, and say which evidence caused the change. (b) Draw and label a particle-level sketch comparing the ions in lemon juice with the ions in Lake Magadi water. (c) Answer the driving question in two to three sentences: why do these substances react so differently with metals, food and living things?	an acid. After testing 12 substances with universal indicator in Lesson 2, I found that NaCl solution is neutral (pH 7), while Lake Magadi water and wood-ash solution are basic (pH 9–11) and also feel harsh. Salts can be neutral, basic or acidic in their effects, and caustic does not mean acidic. (Any change supported by specific evidence from the lessons is acceptable.) (b) Sketch: two beakers. The lemon juice beaker (pH 2–3) is labelled with many H^+ ions and few OH^- ions. The Lake Magadi water beaker (pH about 10.5) is labelled with many OH^- ions and few H^+ ions, with Na^+ ions present from the dissolved salts. Each beaker has a pH label and the word acid or base. (c) Substances behave differently because of the ions they release in water and what those ions meet. Acids such as lemon juice release H^+ ions, which react with metals like aluminium to form a salt and hydrogen gas. Bases such as trona release OH^- ions, which attack skin but make useful soda ash. Salts have different ionic properties, such as how strongly they attract water, so they behave differently in air and in different uses. Knowing this lets us use acids, bases and salts safely, in the right amounts, and protect places like Lake Victoria.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the investigations	Cites specific results from the lessons (pH ranges, Mg + HCl observations, petri dish results, nitrate case study) for every claim.	Cites some specific evidence but some claims rely on general statements.	Gives little or no evidence from the investigations, or evidence is inaccurate.
Particle-level reasoning (ions)	Consistently explains phenomena using H^+ , OH^- , Na^+ , Cl^- and ion–water attraction, accurately linked to pH, reactions and salt behaviour.	Uses ions in most explanations, with minor gaps or errors.	Describes only what is seen, with little or incorrect use of ions.
Chemical vocabulary and equations	Uses terms such as hygroscopic, deliquescent, efflorescent, neutralisation and water of crystallisation correctly, and writes accurate word equations.	Uses most terms correctly; word equations are mostly correct.	Terms are missing or misused; equations are missing or incorrect.
Connection to Kenyan contexts and the anchoring phenomenon	Fully explains both Lake Magadi and the sufuria, and links salt applications and Lake Victoria impacts to the driving question.	Explains both images, with limited links to applications or the environment.	Explains only one image or gives disconnected examples.
Reflection and model revision	Clearly identifies a changed idea, the evidence that changed it, and gives an accurate labelled particle sketch and a coherent answer to the driving question.	Identifies a changed idea with some evidence; sketch and final answer are mostly accurate.	Reflection is vague; sketch or final answer is missing or incorrect.

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